Aleksei Nemtyrev

Statistical Modelling · Causal Inference · Data Science

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Profile

Quantitative researcher finishing a PhD in econometrics (2027). My work is about separating signal from noise: I design estimators that cut variance without losing what matters, and evaluate policy rules that could make the economy less volatile. I build them in Python, R and MATLAB and teach Data Science Methods at MSc level. Before my PhD, I ran a retail company, taking it from break-even to profitability within a year.

Experience

PhD Candidate Aug 2023 – present
Tilburg University, Econometrics & Operations Research Tilburg, NL

- Develop statistical estimators end to end: theoretical derivation, Monte Carlo evaluation, public reproducible code.
- Run simulations on an HPC cluster (MATLAB Parallel Computing Toolbox): 20M+ bootstrap replications per study.
- Develop macroeconomic (DSGE) models and take them to data, from monetary policy rules to model testing.
- Talks at IAAE 2026, CFE-CMStatistics 2026, GIFT 2026, NESG 2024 & 2026, Econometric Society Winter Meeting 2025; co-authors at NBER-NSF 2026, $(EC)^2$ 2026, Chicago Fed 2025, Purdue 2024, Erasmus 2024.
- Teach *Data Science Methods* to 120 MSc students a year (in R: PCA, ridge/lasso, trees, double ML); designed my own model-selection and PCA assignments. Also teach *Statistics 2* to 300 BSc students a year.

General Manager Jun 2016 – Jul 2020
NEVA Torg (retail) St. Petersburg, RU

- Ran operations and finances with a team of 10; grew the chain to seven stores (opened three, closed one).
- Redesigned the stores and moved staff pay to a revenue share: revenue per store rose 30% in 2018.
- Renegotiated supplier contracts, lowering the cost of goods by 15% on average.
- Took the company from break-even in 2017 to RUB 2.5m (≈ EUR 35k) after-tax profit in 2018 (∼9% net margin).

Selected Projects

Targeted Local Projections paper code with O. Boldea · working paper, 2026
Optimally combines local projections and VARs, trading a little bias for ∼20% tighter long-horizon confidence intervals.

FOMC Responses to Credit Spreads with J. R. Campbell · in progress
Proposes adding credit spreads to the Fed's policy rule; in SVAR simulations, this cuts unemployment volatility by 30%.

Corrected Local Projections for Structural IRFs with O. Boldea, A. Cornea-Madeira, A. R. Hall · in progress
Proposes an estimator that strips predictable noise out of local projections; theory and simulations show lower variance.

DSGE Identification and Testing with J. R. Campbell · in progress
Novel utility function and model simplification make a macro model estimable and testable directly from data.

Education

PhD, Econometrics, *Tilburg University* 2023 – 2027 (expected)
Research Master, Economics, *Tilburg University · CentER Scholarship* 2021 – 2023
MSc Economics, Monetary Policy & Banking, *University of Amsterdam · Amsterdam Merit Scholarship* 2020 – 2021
MA (Hons) Economics, *University of Glasgow* 2012 – 2016

Skills

Machine learning	Bias–variance trade-off, ridge, lasso, cross-validation, PCA, classification, random forests, boosting
Causal inference	Difference-in-differences, IV, GMM, double ML, structural VARs, local projections, impulse responses
Time series	Forecasting (VARs, out-of-sample evaluation), ARMA, state-space models, Kalman filter
Statistics	Bayesian inference (own MCMC sampler in Python), MLE, bootstrap, Monte Carlo simulation
Programming	Python (NumPy, SciPy, statsmodels, matplotlib), R, MATLAB, SQL, Stata, LaTeX, Git
AI workflows	Agentic AI (Claude Code, Codex): research codebases, simulation pipelines, document automation
Domain	Macro-finance, monetary policy, credit spreads, FRED data, DSGE and macro–climate models
Communication	Data visualisation, research writing, presenting technical results to expert and general audiences
Business	General management, team leadership, accounting, supplier management, retail operations
Languages	English (C2), Dutch (B1), Russian (native)